

Notes: Cornaggia & Cornaggia (RFS, 2013)

Estimating the Costs of Issuer-Paid Credit Ratings

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Two rating systems	2
2.2	Sample construction	3
2.3	Rating scale mappings	3
2.4	Key outcome variables	3
3	Models and empirical specifications	4
3.1	Rating relation over bond life	4
3.2	Reversal-rate calculation	4
3.3	Timeliness gap	4
3.4	Loss avoidance	4
3.5	True positives, false positives, and false negatives	4
3.6	Accuracy ratios	5
4	Identification and empirical design	5
4.1	What identifies the comparison	5
4.2	Why same-bond comparisons matter	5
4.3	Why the calibrated FHR test matters	5
4.4	Limitations	6
5	Tables and figures	7
5.1	Table 1 and Table 2: rating distributions and coverage differences	7
5.2	Table 3 and Figure 1: issue-level versus firm-level ratings and disagreement	7
5.3	Table 4 and Figure 2: maturity dynamics and default activity	8
5.4	Figure 3 and Table 5: reversal rates and ratings volatility	8
5.5	Figure 4 and Figure 5: false positives versus delayed default warnings	9
5.6	Table 6: advance notice before default	10
5.7	Table 7: loss avoidance from earlier downgrades	11
5.8	Table 8 and Table 9: calibrated scale and accuracy ratios	12
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** What are the costs of relying on issuer-paid credit ratings when ratings agencies trade off *stability* against *timeliness*?
- **Core comparison.** The paper compares Moody’s Investors Service, a traditional issuer-paid credit rating agency, with Rapid Ratings, an independent subscriber-paid rater that produces Financial Health Ratings (FHRs).
- **Main tension.** Timely downgrades help investors avoid losses before default, but premature downgrades can hurt issuers and investors by triggering forced sales, recontracting, regulatory consequences, and unnecessary portfolio turnover.
- **Moody’s policy.** Moody’s appears to favor ratings stability and to “rate through” the economic cycle. This means fewer reversals and fewer false positives, but slower identification of default risk.
- **Rapid Ratings policy.** Rapid Ratings’ FHRs are more volatile and more likely to reverse, but they move faster as fundamentals deteriorate.
- **Key empirical result.** Rapid Ratings downgrades defaulting issuers to speculative grade earlier than Moody’s downgrades the corresponding bonds. The average timeliness gap is about 315 days for all issuers and about 455 days for industrials.
- **Loss avoidance result.** Defaulting bonds trade at higher prices when Rapid Ratings sends the downgrade signal than when Moody’s sends it. In the baseline sample, prices are 66.16% of par at the Rapid Ratings downgrade versus 59.64% at the Moody’s downgrade; among industrials, 96.24% versus 83.22%.
- **Stability cost.** Rapid Ratings generates more reversals and more downgrades of nondefaulting firms, so a pure speed criterion is not enough.
- **Accuracy result.** Accuracy ratios generally show that Rapid Ratings ranks default risk better than Moody’s, especially at longer horizons.
- **Economic takeaway.** The issuer-paid system’s preference for stability reduces noisy false alarms, but the delay in identifying deteriorating credit risk creates economically meaningful losses for investors and other users of ratings-based contracts.

2 Data and measurement

2.1 Two rating systems

- **Moody’s Credit Ratings (MCRs).** Moody’s assigns issue-level alphanumeric ratings such as Aaa, Aa, A, Baa, Ba, B, Caa, Ca, and C.
- **Rapid Ratings Financial Health Ratings (FHRs).** Rapid Ratings assigns firm-level numeric ratings on a 1–100 scale.
- Moody’s is issuer-paid; Rapid Ratings is subscriber-paid.
- The paper treats Moody’s as representative of the Big 3 rating agencies and Rapid Ratings as an independent alternative benchmark.

Interpretation: Moody’s and Rapid Ratings differ not only in compensation model but also in rating technology. Moody’s uses analyst judgment and a stability policy; Rapid Ratings uses a more quantitative, frequently updated firm-level system.

2.2 Sample construction

- The sample covers firms with debt obligations issued between 1988 and 2009.
- Moody’s ratings come from Moody’s Investors Service.
- Rapid Ratings provides proprietary FHRs.
- The main intersection sample contains 2,203 issuers covered by both Moody’s and Rapid Ratings.
- The authors often report results for all issuers and separately for industrial firms.
- Bond prices are used for loss-avoidance tests, with TRACE data used when available.

Interpretation: The empirical design is a within-issuer comparison of two rating technologies and incentive systems, not a randomized experiment.

2.3 Rating scale mappings

The paper uses several mappings across Moody’s and Rapid Ratings scales.

- Moody’s 21-point scale is assigned increasing numerical values, with C at the bottom and Aaa at the top.
- For broad comparisons, Moody’s alphanumeric ratings are mapped into nine categories.
- Rapid Ratings’ FHRs are also mapped into nine broad categories.
- The standard Rapid Ratings investment-grade cutoff is:

$$FHR \geq 65.$$

- The standard Moody’s investment-grade cutoff is Baa3 or above.
- In a later robustness exercise, the paper calibrates Rapid Ratings’ scale to mimic Moody’s downgrade and reversal frequencies, using:

$$FHR_{high} = 59, \quad FHR_{low} = 32.$$

Interpretation: The scale mapping matters because a raw comparison between a 21-notch issue-level scale and a 100-point firm-level scale could mechanically favor one rater. The calibrated thresholds are designed to ask whether Rapid Ratings remains faster even after being forced to behave more like Moody’s.

2.4 Key outcome variables

- **Disagreement:** whether Moody’s and Rapid Ratings assign the same broad credit-risk category.
- **Reversal:** a rating crosses from investment grade to speculative grade and then returns to investment grade within one year or five years.
- **Advance notice:** number of days before default that a rater first moves a defaulting issuer or bond into speculative grade.
- **Loss avoidance:** bond price at the rating downgrade date, interpreted as how much value could be preserved by acting on the signal.
- **Accuracy ratio:** ordinal ranking measure comparing how well ratings rank future default risk.

Interpretation: The paper intentionally separates false positives from false negatives. A fast rater is not necessarily better if speed simply produces noisy downgrades.

3 Models and empirical specifications

3.1 Rating relation over bond life

A key regression relates Rapid Ratings' FHR to Moody's credit rating at issuance and after five years:

$$FHR_{i,t} = \alpha + \beta_1 MCR_{i,t} + \beta_2 MCR_{i,t}^2 + \gamma_i + \varepsilon_{i,t}. \quad (1)$$

- $FHR_{i,t}$ is the Rapid Ratings score associated with the issuer of bond i .
- $MCR_{i,t}$ is the numeric Moody's rating.
- MCR^2 allows nonlinearities in the mapping.
- γ_i are issuer fixed effects in some specifications.

Interpretation: If the relationship weakens after issuance, Moody's and Rapid Ratings are not simply different labels for the same stable credit-risk signal. The divergence suggests different surveillance and updating behavior.

3.2 Reversal-rate calculation

For rater r , horizon $h \in \{1, 5\}$, and year t , the reversal rate is

$$RevRate_{h,t}^r = \frac{\#\{\text{downgrades that reverse within } h \text{ years}\}}{\#\{\text{rating observations or rating actions}\}}. \quad (2)$$

Interpretation: A high reversal rate is evidence of rating volatility or false alarms. A low reversal rate can be good, but it may also reflect delayed downgrades that avoid reversals by waiting too long.

3.3 Timeliness gap

For a defaulting bond i , define

$$Gap_i = Days_i^{FHR} - Days_i^{MCR}, \quad (3)$$

where $Days_i^{FHR}$ is the number of days before default that Rapid Ratings moves the issuer into speculative territory and $Days_i^{MCR}$ is the number of days before default that Moody's moves the bond into speculative territory.

Interpretation: A positive gap means Rapid Ratings gives earlier warning. The paper finds a large positive average gap for defaulting issuers.

3.4 Loss avoidance

Let P_i^{FHR} be the bond price at the Rapid Ratings downgrade and P_i^{MCR} be the bond price at the Moody's downgrade. The loss-avoidance price gap is

$$\Delta P_i = P_i^{FHR} - P_i^{MCR}. \quad (4)$$

Interpretation: If $\Delta P_i > 0$, an investor who sells at the Rapid Ratings downgrade exits at a higher price than an investor who waits for the Moody's downgrade.

3.5 True positives, false positives, and false negatives

- A **true positive** is a downgrade to speculative grade of a bond that later defaults.
- A **false positive** is a downgrade of a bond that does not default.
- A **false negative** can occur when the rating system fails to identify default risk, or upgrades a bond that later defaults.

Interpretation: The paper’s cost calculation weighs the benefit of earlier true-positive downgrades against the costs of false-positive sales and rating volatility.

3.6 Accuracy ratios

The accuracy ratio measures whether a rater assigns worse ratings to bonds that subsequently default than to bonds that do not default. Conceptually,

$$AR = \text{ranking quality of rating scores for future default outcomes.} \quad (5)$$

Interpretation: A higher accuracy ratio means the rater better sorts issuers by default risk. This is an ordinal criterion: it evaluates ranking, not only calibration.

4 Identification and empirical design

4.1 What identifies the comparison

- The main evidence comes from comparing Moody’s and Rapid Ratings for overlapping issuers and bonds.
- The paper tracks how both ratings evolve over time, especially around downgrades and defaults.
- Defaulting bonds provide a natural benchmark for timeliness: a rating system should identify deterioration before default.
- Nondefaulting downgrades provide a benchmark for stability: a rating system should not trigger too many false alarms.

Interpretation: The analysis uses a horse race between two rating systems with different compensation structures and updating policies.

4.2 Why same-bond comparisons matter

- Bond-level comparisons reduce the concern that one rater covers intrinsically different firms.
- The authors examine all issuers and industrials separately because financial and conduit issuers can have different rating dynamics.
- The paper also handles the issue that Moody’s ratings are bond-level while FHRs are firm-level.

Interpretation: The firm-level versus issue-level distinction could be a confound, but Table 3 shows that within-firm-year Moody’s ratings are usually not very dispersed.

4.3 Why the calibrated FHR test matters

- Rapid Ratings is more volatile than Moody’s.
- A skeptic could argue that Rapid Ratings only looks better because it downgrades too many firms.
- The calibration forces Rapid Ratings to have downgrade and reversal frequencies closer to Moody’s.
- Rapid Ratings still provides earlier signals and better loss avoidance after this calibration.

Interpretation: This is one of the paper’s most important robustness ideas: even after making Rapid Ratings behave more like Moody’s, Moody’s remains slower.

4.4 Limitations

- Moody's and Rapid Ratings differ in both compensation model and technology, so the paper cannot isolate compensation alone.
- Rapid Ratings' FHR is firm-level, while Moody's MCR is issue-level.
- The analysis is not randomized; coverage and rating actions are selected.
- The loss-avoidance calculation focuses on bond investors and does not fully quantify issuer costs, customer effects, supplier effects, or macroeconomic effects.

Interpretation: The evidence strongly suggests that issuer-paid ratings are slower, but the paper is careful not to claim a complete social-welfare calculation.

5 Tables and figures

5.1 Table 1 and Table 2: rating distributions and coverage differences

Read:

- Table 1 shows that the average Moody's rating for all issuers is about A2, while the industrial average is about Ba1.
- FHRs are generally lower for industrials than for all issuers, and FHRs attached to bonds are higher for all issuers than issuer-level FHRs.
- Table 2 shows that firms covered by both raters are larger than firms covered by only one rater.
- Rapid-Ratings-only firms tend to be smaller and have lower leverage because many lack public bonds.

Economic message: The raters cover related but not identical populations. The overlapping sample is necessary for the paper's main comparison.

Table 1
Summary statistics of Moody's credit ratings and Rapid Ratings' Financial Health Ratings

	FHR		FHR attached to bonds		MCR	
	All issuers	Industrials	All issuers	Industrials	All issuers	Industrials
Mean	55.15	53.64	65.93	55.90	16.0 (≈A2)	11.0 (≈Ba1)
SD	20.21	20.86	16.99	20.14	5.1	4.3
Minimum	5	5	5	5	1 (=C)	1 (=C)
5th percentile	17	16	29	17	6 (=B3)	4 (=Caa2)
25th percentile	41	38	59	42	13 (=Baa2)	8 (=B1)
Median	60	58	71	62	17 (=A1)	11 (=Ba1)
75th percentile	71	71	79	72	21 (=Aaa)	14 (=Baa1)
95th percentile	81	81	84	81	21 (=Aaa)	18 (=Aa3)
Maximum	94	94	94	94	21 (=Aaa)	21 (=Aaa)
N ratings	58,396	46,425	947,538	242,656	142,324	35,293

This table displays summary statistics for credit ratings assigned by Moody's and FHRs assigned by Rapid Ratings. Each observation represents the assignment of a new rating. Moody's assigns its credit rating to individual bonds and Rapid Ratings assigns its FHRs to firms. We also compute summary statistics after associating the firm-level FHRs with all of a given firm's outstanding bonds. The sample includes ratings assigned to firms with debt obligations issued between 1988 and 2009. Moody's credit ratings come from Moody's Investors Service and Rapid Ratings provides its FHRs.

Table 2
Comparison of firm characteristics by rating agency(ies) providing coverage

	All issuers, ratings provided by						Industrials, ratings provided by					
	Moody's (1)	Moody's and RR (2)	RR (3)	(2)-(1) (p-value)	(2)-(3) (p-value)	(1)-(3) (p-value)	Moody's (4)	Moody's and RR (5)	RR (6)	(5)-(4) (p-value)	(5)-(6) (p-value)	(4)-(6) (p-value)
Total assets (\$mil)												
N firm-years	12,039	25,332	40,186				8,277	17,408	19,130			
Mean	12,546	16,083	1,896	5.539**	16,189**	10,650**	2,215	7,735	1,483	2,521**	6,252**	3,732**
Median	1,087	1,972	200	(0.000)	(0.000)	(0.000)	758	1,678	191	(0.000)	(0.000)	(0.000)
SD	70,024	110,923	25,046				18,601	26,774	11,080			
Leverage (%)												
N firm-years	11,909	25,232	39,928				8,236	17,365	19,052			
Mean	31	32	22	0.93	9.90**	8.97**	34	33	19	-1.87	13.07**	14.94**
Median	28	28	10	(0.449)	(0.000)	(0.000)	30	29	11	(0.270)	(0.000)	(0.011)
SD	182	50	379				218	34	49			
Revenue (\$mil)												
N firm-years	11,957	25,272	40,019				8,253	17,388	19,102			
Mean	2,696	6,480	857	3.784**	5,625**	1,839**	1,261	6,491	1,225	3,231**	5,265**	2,035**
Median	516	1,406	127	(0.000)	(0.000)	(0.000)	648	1,595	180	(0.000)	(0.000)	(0.000)
SD	8,169	17,799	5,631				9,092	18,315	7,775			
Net income (\$mil)												
N firm-years	11,988	25,303	40,098				8,256	17,388	19,101			
Mean	151	379	38	219**	332**	113**	145	359	55	214**	304**	90**
Median	28	54	18	(0.000)	(0.000)	(0.000)	18	55	6	(0.000)	(0.000)	(0.000)
SD	1,232	1,791	795				1,066	1,779	1,108			

This table displays firm-year summary statistics for characteristics of firms receiving credit ratings. The characteristics include Total assets, Leverage (the ratio of Long-Term Debt and Debt to Current Liabilities to Total Assets), Revenue, and Net income. The data come from Compustat. Columns 1 and 4 summarize these characteristics for firms that receive a credit rating only from Moody's. Columns 2 and 5 summarize these characteristics for firms that receive ratings from both Moody's and Rapid Ratings. Columns 3 and 6 summarize these characteristics for firms that receive ratings only from Rapid Ratings. We use asterisks to indicate significant differences between Moody's and Rapid Ratings at the 1% level. ** indicates significance at the 5% level. *** indicates significance at the 1% level.

5.2 Table 3 and Figure 1: issue-level versus firm-level ratings and disagreement

Read:

- Table 3 shows that 54% of firm-years have more than one bond outstanding, but the median firm has only two bonds and a zero notch spread.
- The average Moody's notch spread within firm-year is only 0.27, so issue-level dispersion is usually modest.
- Figure 1 shows substantial disagreement between Moody's and Rapid Ratings over time.
- Moody's is often higher than Rapid Ratings, but the degree of disagreement changes over the cycle.

Economic message: The firm-level nature of Rapid Ratings is not fatal to the comparison because most firms' Moody's issue-level ratings are clustered closely. The disagreement reflects real differences in updating and surveillance.

Moody's credit ratings within firm-years

Panel A: Number of bonds outstanding and distributions of Moody's credit ratings within firm-years

	N firm-years	Mean	Minimum	5%	Median	95%	Maximum
Multiple bonds outstanding indicator	9,112	0.54	0	0	1	1	1
Number of bonds outstanding	9,112	10.83	1	1	2	18	2,034
Nitch Spread	9,112	0.27	0	0	0	2	15

Panel B: Descriptions of highest- and lowest-rated bonds for firm-years with multiple bonds outstanding

	Highest-rated bond within firm-year					Lowest-rated bond within firm-year					
	N bond-years	Mean	5%	Median	95%	N bond-years	Mean	5%	Median	95%	Difference in means
Moody's credit rating	4,963	12.9	5	13	20	4,963	12.4	5	13	19	-0.3***
	(=Baa2)	(=Caa1)	(=Baa2)	(=Aaa1)	(=Baa3)		(=Baa2)	(=Caa1)	(=Baa2)	(=Aaa2)	
Face value	4,829	303	15	200	1,000	4,848	313	23	200	1,000	-10
Coupon rate	4,431	7.35	5.00	7.25	11.50	4,487	7.49	3.50	7.31	11.75	-0.14***
Years until maturity	4,872	11.9	2.1	10.0	30.0	4,803	12.7	3.0	10.0	30.0	-0.8***

Panel A displays summary statistics for the number of bonds outstanding within firm-years and the distribution of Moody's credit ratings within firm-years. Multiple bonds outstanding indicator takes a value of one if a firm has more than one bond outstanding as of January 1 of a given year and is zero if the firm has only one bond outstanding at that time. Number of bonds outstanding represents the number of bonds outstanding for a given firm and a given year. Nitch spread represents the difference between the highest- and lowest-rated outstanding bond for a given firm and a given year, measured in credit rating notches. Nitch spread takes a value of zero if the firm has only one bond outstanding. Panel B displays characteristics of the highest- and lowest-rated bonds belonging to firms with more than one bond outstanding. Face value is the face value of the bond expressed in millions of U.S. dollars. Coupon rate is the coupon rate expressed as a percentage of par. Years to maturity is the number of years remaining until the bond matures. Face values, coupon rates, and years until maturity come from Moody's Default and Recovery Database. The sample, which runs from 1988 to 2009, includes firms with FHRs assigned by Rapid Ratings and bonds rated by Moody's Investors Service. *, **, and *** indicate a significant difference at the 10%, 5%, or 1% level, respectively. We cluster standard errors at the firm level.

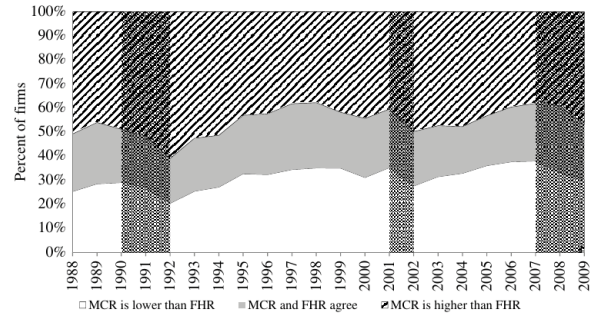


Figure 1
Distribution of agreement between Moody's and Rapid Ratings

5.3 Table 4 and Figure 2: maturity dynamics and default activity

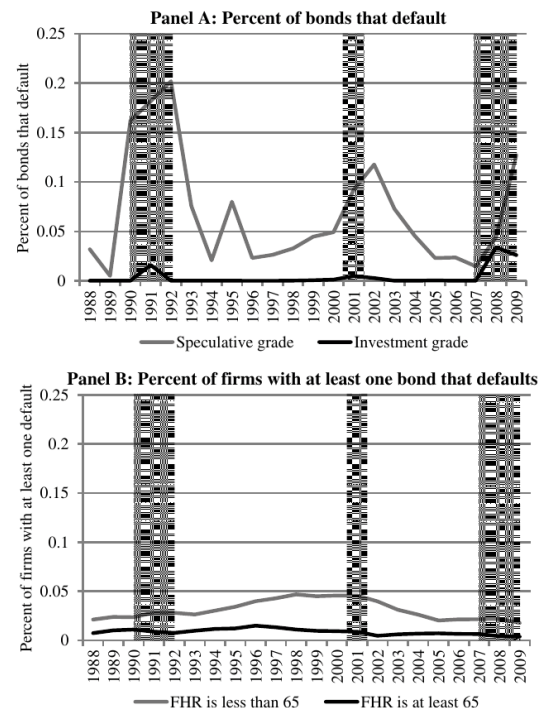
Read:

- Table 4 relates FHRs to Moody's ratings at issuance and five years later.
- The relationship remains positive, but explanatory power falls and RMSE rises as bonds mature.
- With issuer fixed effects, the five-year relation is weaker than the issuance relation.
- Figure 2 shows that default activity varies strongly over time and is concentrated in speculative-grade buckets.

Economic message: Moody's and Rapid Ratings do not simply produce equivalent signals that stay aligned over time. Their signals diverge as credit conditions evolve.

	FHR at issue (1)	FHR 5 yrs after issue (2)	FHR at issue (3)	FHR 5 yrs after issue (4)	FHR at issue (5)	FHR 5 yrs after issue (6)	FHR at issue (7)	FHR 5 yrs after issue (8)
MCR at issue	2.4917*** (0.1380)		0.9803 (0.7980)		1.5286*** (0.3620)		5.6609*** (0.9204)	-0.1470*** (0.0344)
MCR ² at issue		0.0632** (0.0270)						
MCR 5 years after issue		2.2061*** (0.1866)		3.4544*** (0.8042)		1.3150*** (0.3161)		2.5993** (1.2016)
MCR ² 5 years after issue				-0.0430 (0.0276)				-0.0668 (0.0465)
Constant	27.1006*** (2.2917)	27.1962*** (3.6145)	39.6219*** (4.5590)	44.2923*** (6.4821)	42.1432*** (5.3025)	19.8651*** (6.7183)	35.7538*** (7.5731)	35.7538*** (7.5731)
Issuer fixed effects?	no	no	no	yes	yes	yes	yes	yes
N	26,517	14,269	26,517	14,269	26,517	14,269	14,269	14,269
R ²	0.4562	0.3320	0.4826	0.3352	0.8116	0.6551	0.8148	0.6554
R ² homotyped SE	(0.0966)	(0.0600)	(0.0970)	(0.0610)	(0.0307)	(0.0623)	(0.0310)	(0.0310)
RMSE	11.6140	10.0440	11.3940	10.0130	6.9260	10.4388	6.9678	10.4542
RMSE homotyped SE	(1.9434)	(0.7720)	(1.9310)	(0.8090)	(1.1062)	(0.6361)	(1.1090)	(0.6509)
		(2) - (1)		(4) - (3)		(6) - (5)		(8) - (7)
R ² difference		-0.1242		-0.1276		-0.1565**		-0.1504**
Std. Error		(0.1177)		(0.1142)		(0.0790)		(0.0728)
RMSE difference		2.5830		2.6190		3.5728***		3.5864***
SE		(2.0913)		(2.0914)		(1.2760)		(1.2902)

This table displays regressions with Rapid Ratings' FHRs (we assign to each bond the FHR of its issuing firm) as the dependent variable and Moody's credit ratings (and Moody's credit ratings squared) as the independent variable. We perform this regression using observations at the time bonds are issued and after the bonds have matured five years. Standard errors appear in parentheses.



5.4 Figure 3 and Table 5: reversal rates and ratings volatility

Read:

- Figure 3 shows that default activity varies strongly over time and is concentrated in speculative-grade buckets.

- Rapid Ratings' one-year reversal rate among industrials ranges roughly from 0.5% to 2.5%; its five-year reversal rate is higher still.
- Table 5 confirms the reversal-rate difference across recessions, expansions, and rating categories.
- Moody's reversal rates are very low; only some five-year industrial measures approach 1%.

Economic message: Moody's stability has value. It avoids many premature downgrades and repeated crossings of the investment-grade boundary.

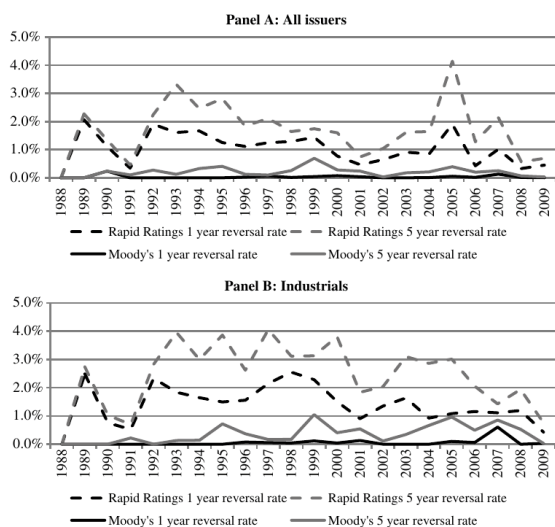


Table 5

Reversals

Panel A: Reversals by economic conditions at the time of reversal

	Rapid Ratings				Moody's			
	N within 1 year of downgrade	N within 5 years of downgrade	N ratings	1-year reversal rate	5-year reversal rate	N within 1 year of downgrade	N within 5 years of downgrade	N ratings
All issuers	1,097	1,730	237,136	0.46%	0.74%	14	44	61,365
Recessions	8,058	14,905	806,928	1.00%	1.85%	38	305	137,266
Expansions	466	817	476,96	0.98%	1.71%	4	23	8,340
Industrials	3,636	6,079	240,912	1.51%	2.77%	30	177	36,418

(continued)

Continued

Panel B: Reversals by bonds' credit ratings at issuance

	Rapid Ratings				Moody's			
	N within 1 year of downgrade	N within 5 years of downgrade	N ratings	1-year reversal rate	5-year reversal rate	N within 1 year of downgrade	N within 5 years of downgrade	N ratings
All issuers, Rapid Ratings defines its rating categories								
65-70	3,465	5,182	161,797	2.14%	3.20%			
71-80	2,166	3,681	312,466	0.69%	1.18%			
81-100	359	954	274,583	0.13%	0.35%			
All issuers, rating categories defined by matching to Moody's ratings distribution at issue								
40-49	2,861	3,496	51,752	5.53%	6.76%			
50-66	2,665	3,844	218,454	1.22%	1.76%			
67-100	8,410	9,186	703,623	1.20%	1.31%			
Industrials, Rapid Ratings defines its rating categories								
65-70	1,308	1,999	50,803	2.57%	3.93%			
71-80	1,028	1,687	62,287	1.65%	2.71%			
81-100	133	263	13,612	0.98%	1.93%			
Industrials, rating categories defined by matching to Moody's ratings distribution at issue								
44-64	1,489	3,016	93,373	1.59%	3.23%			
65-77	898	1,569	98,173	0.91%	1.60%			
78-100	147	236	28,529	0.52%	0.83%			

This table displays counts and probabilities of instances in which bonds' ratings migrate into investment grade after having migrated to speculative grade in the past one or five years. Panel A displays ratings reversals according to the economic conditions at the time of the reversal. To calculate Moody's one-year (five-year) reversal rate, we count the number of ratings assigned by Moody's that represent a reversal within one year (five years) of a downgrade and that occur during a recession (or, separately, an expansion). We divide this number by the total number of ratings assigned by Moody's during recessions in the sample period (1988 to 2009). To calculate Rapid Ratings' one-year (five-year) reversal rate, we count the number of bonds issued by firms with FHRs that reverse within one year (five years) of a downgrade and that occur during a recession (or, separately, an expansion). (An FHR of at least 65 is investment grade on

5.5 Figure 4 and Figure 5: false positives versus delayed default warnings

Read:

- Figure 4 plots bond prices around FHR downgrades for firms that do not default and that Moody's does not downgrade.
- The event study illustrates the cost of acting on an FHR downgrade that later appears premature.
- Figure 5 focuses on defaulting bonds. Moody's ratings stay relatively high until close to default, while FHRs and z-scores deteriorate earlier.

- Bond prices decline well before Moody's downgrade, suggesting that market prices reflect deterioration before Moody's updates the rating.

Economic message: The paper's main tradeoff is visible here: Rapid Ratings can generate costly false positives, but Moody's can be dangerously late for true distress.

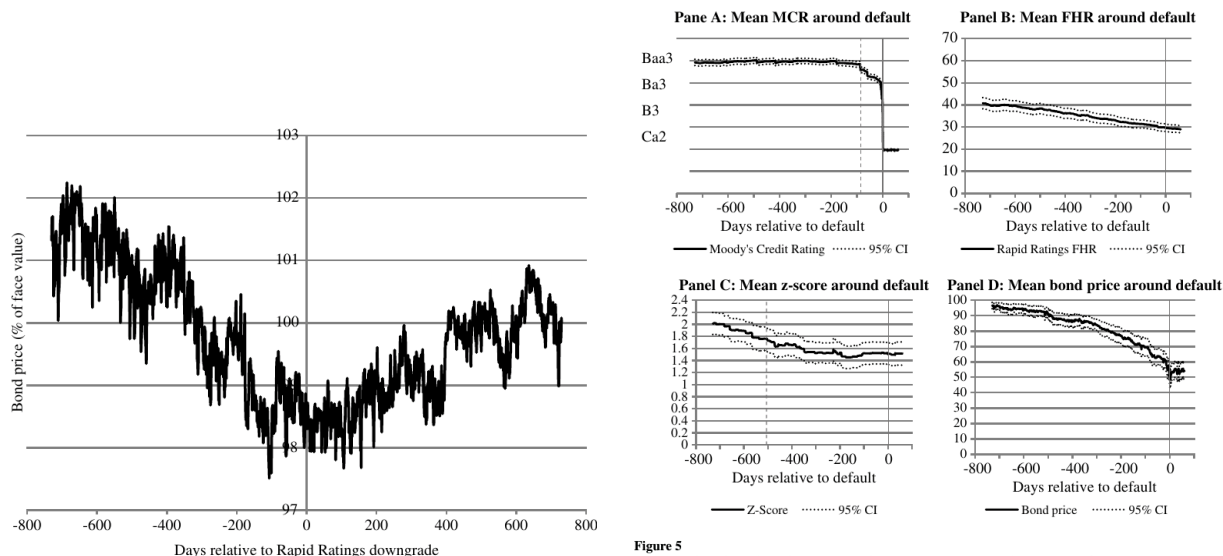


Figure 4

Figure 5
Mean credit rating, Financial Health Rating, z-score, and bond price around default
This figure plots mean Moody's credit ratings, Rapid Ratings' FHRs, z-scores, and bond prices, expressed as a percentage of face value, from two years prior to default (-730 days) to two months after default (+60 days).

5.6 Table 6: advance notice before default

Read:

- For all issuers, Rapid Ratings downgrades defaulting issuers an average of 793 days before default versus 478 days for Moody's, a 315-day difference.
- The median gap is also large: 473 days for Rapid Ratings versus 7 days for Moody's.
- For industrials, the average gap is 455 days, and both mean and median comparisons favor Rapid Ratings.
- The largest all-issuer gap occurs in recessions, where Rapid Ratings is more than 1,000 days earlier on average.

Economic message: Moody's stable ratings come at the cost of delayed default-risk recognition. For investors constrained by ratings, this delay can be very expensive.

Table 6

Advance notice

Panel A: Defaults by all issuers

	Moody's downgrades (1)	Rapid Ratings downgrades (2)	Diff. in mean (2) – (1) (SE)	Diff. in median z-score (p-value)
Mean	478	793	315***	21.9700***
Median	7	473	(43.20)	(0.0000)
SD	1,086	996		
N bonds	1,601	878		
N issuing firms	122	128		

Defaults split by economic conditions at time of default

Defaults in recessions				
Mean	144	1,162	1,018***	12.1880***
Median	0	778	(87.20)	(0.0000)
SD	465	1,180		
N bonds	923	189		
N issuing firms	44	45		
Default sin expansions				
Mean	933	692	–241***	9.6970***
Median	87	460	(63.60)	(0.0000)
SD	1,461	786		
N bonds	678	689		
N issuing firms	78	83		

Defaults split by credit ratings at issuance

Bonds rated Aa3, Aa2, Aa1, or Aaa by Moody's or 67–100 by Rapid Ratings				
Mean	493	670	177***	14.0360***
Median	0	547	(48.58)	(0.0000)
SD	1,293	666		
N bonds	1,336	400		
N issuing firms	38	92		
Bonds rated A3, A2, or A1 by Moody's or 50–66 by Rapid Ratings				
Mean	398	1,022	624***	16.0730***
Median	0	838	(59.0057)	(0.0000)
SD	1,159	996		
N bonds	1,309	404		
N issuing firms	45	125		
Bonds rated Baa3, Baa2, or Baa1 by Moody's or 40–49 by Rapid Ratings				
Mean	1,016	771	–245**	3.0180***
Median	447	558	(98.13)	(0.0025)
SD	1,411	774		
N bonds	263	291		
N issuing firms	51	107		

Panel B: Defaults by industrials

	Moody's downgrades (1)	Rapid Ratings downgrades (2)	Diff. in mean (2) – (1) (SE)	Diff. in median z-score (p-value)
Mean	1,011	1,465	455***	3.3930***
Median	970	1,034	(100.16)	(0.0007)
SD	865	1,345		
N bonds	250	257		
N issuing firms	94	96		

Defaults split by economic conditions at time of default

Defaults in recessions				
Mean	1,428	2,004	576***	0.2980
Median	1,288	1,400	(200.22)	(0.7658)
SD	630	1,731		
N bonds	82	85		
N issuing firms	34	33		

Table 6
Continued

Defaults in expansions				
Mean	807	1,199	392***	4.1230***
Median	554	998	(103.30)	(0.0000)
SD	892	1,011		
N bonds	168	172		
N issuing firms	60	63		

Defaults split by credit ratings at issuance

Bonds rated Aa3, Aa2, Aa1, or Aaa by Moody's or 78–100 by Rapid Ratings				
Mean	1,006	623	–383	–1.3190
Median	606	275	(352.19)	(0.1870)
SD	1,584	1,009		
N bonds	25	43		
N issuing firms	9	19		
Bonds rated A3, A2, or A1 by Moody's or 65–77 by Rapid Ratings				
Mean	962	1,039	77	–1.9340*
Median	971	700	(132.48)	(0.0531)
SD	864	1,061		
N bonds	71	160		
N issuing firms	25	79		
Bonds rated Baa3, Baa2, or Baa1 by Moody's or 44–64 by Rapid Ratings				
Mean	741	1,105	364***	2.6880***
Median	275	768	(102.14)	(0.0072)
SD	980	1,121		
N bonds	144	334		
N issuing firms	40	145		

Panel C: Stock price movement prior to defaults

	Peak stock price	Stock drops 50% off peak first time	Stock drops 50% peak permanently
Mean	3,221	1,698	226
Median	2,968	623	–203
SD	1,684	2,159	846
N bonds	469	445	445
N issuing firms	44	38	38

This table displays summary statistics for the number of days prior to bonds' defaults for several benchmark Moody's downgrades represents the number of days prior to default that the bond's credit rating migrates below the investment-grade threshold. *Rapid Ratings downgrades* represents the number of days prior to default that the FHR of the issuing firm migrates below 65. *Peak stock price* represents the number of days prior to default that the issuing firm's stock price reaches its peak. *Stock drops 50% off peak first time* represents the number of days prior to default that the stock price of the issuing firm drops 50% off its peak value over the life of the bond for the first time. *Stock drops 50% off peak permanently* represents the number of days prior to default that the issuing firm's stock price drops below 50% of its peak value over the life of the bond and never returns above that value. We conduct two-tailed *t*-tests for differences in means, and we report *z*-scores from Wilcoxon rank-sum tests for differences in medians. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level respectively.

Continued

5.7 Table 7: loss avoidance from earlier downgrades

Read:

- Panel A shows defaulting bonds trade at 66.16% of par at the Rapid Ratings downgrade versus 59.64% at the Moody's downgrade.
- For industrials, the contrast is larger: 96.24% of par at the Rapid Ratings downgrade versus 83.22% at the Moody's downgrade.
- Panel B shows especially large price gaps in recessions for all issuers.
- Panel C shows price advantages across initial rating categories, especially for all issuers.

Economic message: Acting on Rapid Ratings' signal can preserve value before default. This is the paper's most direct dollar-cost evidence on delayed issuer-paid ratings.

Table 7
Loss avoidance

Panel A: Bond characteristics on downgrade dates for all defaults and recovery prices after default

	All issuers				Industrials			
	Moody's downgrades (1)	Rapid Ratings downgrades (2)	Diff. in mean (2) – (1) (SE)	Diff. in median z-score (p-value)	Moody's downgrades (3)	Rapid Ratings downgrades (4)	Diff. in mean (4) – (3) (SE)	Diff. in median z-score (p-value)
Bond price								
Mean	59.64	66.16	6.52***	2.412**	83.22	96.24	13.02***	4.288***
Median	57.90	60.00	(1.71)	(0.016)	86.41	99.66	(3.40)	(0.000)
SD	29.75	26.01			18.22	14.48		
N bonds	796	373			41	60		
N issuing firms	22	42			12	30		
Bond YTM								
Mean	32.71	20.14	–12.58***	1.007	11.15	10.32	–0.82	–0.628
Median	17.79	22.11	(3.86)	(0.314)	9.07	7.60	(1.83)	(0.209)
SD	104.08	10.21			7.56	10.07		
N bonds	743	364			39	54		
N issuing firms	20	37			11	27		
Days advance								
Mean	84	503	419***	25.640***	943	1,499	556***	3.394***
Median	0	111	(38.80)	(0.000)	1,171	1,489	(137.2)	(0.001)
SD	251	729			590	787		
N bonds	796	373			41	60		
N issuing firms	22	42			12	30		

Continued

Panel C: Bond prices on downgrade dates for defaults split by credit ratings at issuance using rating categories defined by matching Rapid Ratings to Moody's ratings distribution

	All issuers				Industrials			
	Moody's downgrades (1)	Rapid Ratings downgrades (2)	Ratings (2) – (1) (SE)	Diff. in mean z-score (p-value)	Moody's downgrades (3)	Rapid Ratings downgrades (4)	Ratings (4) – (3) (SE)	Diff. in mean z-score (p-value)
Mean	57.11	61.74	4.63***	5.956***	64.76	87.10	22.34	1.556
Median	53.62	60.81	(2.19)	(0.000)	62.15	84.79	(15.35)	(0.120)
SD	29.26	19.91			20.42	13.89		
N bonds	704	54			4	10		
N issuing firms	13	25			2	6		
Mean	57.28	61.46	4.18***	6.205***	63.41	86.52	23.11*	2.197**
Median	53.17	60.50	(1.10)	(0.000)	62.59	85.13	(13.81)	(0.028)
SD	28.80	22.01			15.22	16.20		
N bonds	734	57			27	44		
N issuing firms	20	28			6	20		
Mean	72.54	91.21	18.67***	2.769***	83.61	89.01	5.40	1.540
Median	84.09	95.18	(5.16)	(0.006)	84.13	95.18	(4.30)	(0.124)
SD	28.96	16.54			18.01	17.65		
N bonds	42	40			7	49		
N issuing firms	16	12			12	27		

This table displays summary statistics for the price (the most recent end-of-day trade price) and yield to maturity of bonds issued with investment-grade ratings that later default. We examine these characteristics when the bonds' credit ratings migrate into speculative-grade territory. Moody's downgrade is the date prior to default that the bond's credit rating migrates below the investment-grade threshold. Rapid Ratings downgrade is the date prior to default that the FRR of the issuing firm migrates below S1. Panel A summarizes these characteristics for all defaults and separately for defaulting bonds issued by industrial firms. Panel A also indicates the number of days prior to default each rate downgrades the bond as well as recovery prices (bond market prices thirty days after default). Panel B displays prices on downgrade dates for defaulting bonds according to the economic conditions at the time of the default and separately for defaulting bonds issued by industrial firms. Panel C displays prices on downgrade dates for defaulting bonds according to the Moody's credit rating assigned to the bond at issuance or the FRR associated with the issuing firm at the time the bond was issued. Panel C also separately displays these prices for defaulting bonds issued by industrial firms. We conduct two-tailed t-tests for

Continued

Panel B: Bond prices on downgrade dates for defaults split by economic conditions at the time of default

	All issuers				Industrials			
	Moody's downgrades (1)	Rapid Ratings downgrades (2)	Diff. in mean (2) – (1) (SE)	Diff. in median z-score (p-value)	Moody's downgrades (3)	Rapid Ratings downgrades (4)	Diff. in mean (4) – (3) (SE)	Diff. in median z-score (p-value)
Mean	62.05	93.99	31.93***	7.330***	86.56	96.55	9.99***	2.244**
Median	60.00	96.27	(2.65)	(0.000)	85.11	96.27	(2.96)	(0.025)
SD	33.51	14.51			9.86	11.18		
N bonds	479	45			20	31		
N issuing firms	6	18			3	15		
Mean	55.98	59.35	3.37*	0.237	79.84	95.90	16.06***	3.897***
Median	48.48	52.00	(2.03)	(0.031)	86.34	101.04	(6.05)	(0.000)
SD	22.58	28.62			23.36	17.53		
N bonds	317	328			21	29		

5.8 Table 8 and Table 9: calibrated scale and accuracy ratios

Read:

- Table 8 recalibrates Rapid Ratings to mimic Moody's downgrade and reversal propensities.
- Even after calibration, Rapid Ratings gives earlier warning: 1,104 days in advance compared with 424 days for Moody's overlap sample and 84 days for Moody's irrespective of Rapid Ratings.
- The calibrated Rapid Ratings downgrade also occurs at higher bond prices than the Moody's downgrade.
- Table 9 shows higher accuracy ratios for Rapid Ratings at most default horizons.
- At five years, Rapid Ratings' accuracy ratio is 0.2254 versus 0.0867 for Moody's among all issuers.

Economic message: Rapid Ratings' superior timeliness is not only due to volatility. Even after making the scale more Moody's-like, it remains more informative about future default risk.

Table 8
Advance notice and loss avoidance after calibrating Rapid Ratings' scale

	Rapid Ratings (1)	Moody's overlap with Rapid Ratings (2)	Moody's irrespective of Rapid Ratings (3)	Diff. in mean (1) – (2) (SE)	Diff. in mean (1) – (3) (SE)	Diff. in median z-score (p-value)	Diff. in median z-score (p-value)
Bond price							
Mean	89.90	72.75	59.67	17.15***	30.23***	4.37***	10.48***
Median	95.65	80.19	58.25	(3.39)	(1.95)	(0.000)	(0.0000)
SD	16.03	23.51	29.77				
N bonds	95	63	798				
N firms	36	16	22				
Bond YTM							
Mean	17.77	18.75	32.73	−0.98	−14.97**	−4.59***	−8.02***
Median	8.44	15.78	17.96	(5.88)	(6.86)	(0.0000)	(0.0000)
SD	54.08	11.16	103.94				
N bonds	90	60	745				
N firms	32	14	20				
Days in advance							
Mean	1,104	424	84	679***	1,020***	4.88***	11.93***
Median	1,158	87	0	(118)	(92)	(0.0000)	(0.0000)
SD	895	590	251				
N bonds	95	63	798				
N firms	36	16	22				

This table displays, on the date each rater changes its ratings from investment grade to speculative grade, the prices, yields to maturity, and number of days prior to default that Rapid Ratings or Moody's downgrades defaulting bonds. We collect end-of-day trade prices and yields to maturity from TRACE. For entry to the sample in Column 1, we require the issuing firm to have an investment-grade FHR from Rapid Ratings at the time the bond was issued. We consider a FHR of at least 59 to be investment grade and a FHR of less than or equal to 32 to be speculative grade on Rapid Ratings' calibrated scale. We measure the characteristics in the table below on the date Rapid Ratings downgrades the issuing firm's FHR to less than or equal to 32. For Column 2, we start with the sample in Column 1 and restrict it to bonds that Moody's certifies as investment grade at issuance. We measure the characteristics on the dates Moody's downgrades the bonds to speculative grade according to Moody's traditional scale. For entry to the sample in Column 3, we require Moody's to rate a bond investment grade at issuance and for the bond to later default. We measure the bonds' prices and yields to maturity on the dates Moody's downgrades the bonds to speculative grade. We conduct two-tailed *t*-tests for differences in means, and we report *p*-values from Wilcoxon rank-sum tests for differences in medians. *, **, and *** indicate

Table 9
Accuracy ratios

Panel A: Accuracy ratios

Default horizon	All issuers			Industrials		
	Moody's (1)	Rapid Ratings (2)	(2) – (1)	Moody's (3)	Rapid Ratings (4)	(4) – (3)
5 years	0.0867	0.2254	0.1388	0.0984	0.2135	0.1151
4 years	0.1275	0.1572	0.0297	0.1445	0.1625	0.0180
3 years	0.1438	0.1860	0.0422	0.1496	0.1959	0.0464
2 years	0.1582	0.2303	0.0721	0.1708	0.2259	0.0551
1 year	0.2140	0.2222	0.0082	0.2369	0.2393	0.0024

Panel B: Accuracy ratios calculated with defaulting bonds' lowest ratings within default horizon windows

Default horizon	All issuers			Industrials		
	Moody's (1)	Rapid Ratings (2)	(2) – (1)	Moody's (3)	Rapid Ratings (4)	(4) – (3)
5–6 years	0.0866	0.2479	0.1613	0.0978	0.2477	0.1499
4–5 year	0.1310	0.2323	0.1013	0.1461	0.2437	0.0977
3–4 years	0.1405	0.2537	0.1132	0.1625	0.2650	0.1025
2–3 years	0.1629	0.2775	0.1146	0.1869	0.2837	0.0968
1–2 years	0.2752	0.2872	0.0120	0.2957	0.3048	0.0091

Panel C: Accuracy ratios calculated with defaulting bonds' highest ratings within default horizon windows

Default horizon	All issuers			Industrials		
	Moody's (1)	Rapid Ratings (2)	(2) – (1)	Moody's (3)	Rapid Ratings (4)	(4) – (3)
5–6 years	0.0819	0.2062	0.1243	0.0933	0.1968	0.1035
4–5 year	0.1245	0.1525	0.0280	0.1413	0.1501	0.0088
3–4 years	0.1423	0.1731	0.0308	0.1639	0.1620	−0.0018
2–3 years	0.1556	0.2115	0.0560	0.1795	0.2038	0.0243
1–2 years	0.2168	0.1504	−0.0664	0.2403	0.1706	−0.0697

Panel A: All bonds; Moody's and Rapid Ratings' accuracy ratios for defaulting bonds; and Panel B: All bonds; Moody's and Rapid Ratings' accuracy ratios for defaulting bonds' lowest ratings within default horizon windows; and Panel C: All bonds; Moody's and Rapid Ratings' accuracy ratios for defaulting bonds' highest ratings within default horizon windows.

6 Short summary

- The paper estimates the costs of relying on issuer-paid credit ratings by comparing Moody's with Rapid Ratings.
- Moody's ratings are more stable and have fewer reversals.
- Rapid Ratings' FHRs are more volatile and create more false positives.
- However, Rapid Ratings identifies default risk much earlier.
- Rapid Ratings downgrades defaulting issuers roughly 315 days before Moody's in the full sample and roughly 455 days before Moody's among industrials.
- Bond prices are higher at Rapid Ratings downgrades than at Moody's downgrades, so earlier signals provide loss avoidance.
- Moody's ratings lag not only Rapid Ratings but also z-scores and market prices around default.
- After calibrating Rapid Ratings to mimic Moody's downgrade and reversal behavior, Rapid Ratings remains faster and more useful.
- Accuracy ratios show that Rapid Ratings generally provides a better ordinal ranking of default risk.
- The paper concludes that Moody's stability bias exceeds what is needed to control rating volatility and is consistent with issuer-paid conflicts of interest.

7 Conclusion

7.1 Core conclusion

Cornaggia and Cornaggia show that issuer-paid ratings are stable but slow. Moody's avoids many premature downgrades, but that stability delays the recognition of default risk. Rapid Ratings is noisier, yet its earlier signals allow investors to avoid losses before default and generally rank credit risk better.

7.2 Economic intuition

Credit ratings are used in contracts, regulations, and portfolios. A downgrade can force costly actions, so stability has value. But if a rating agency waits too long, investors hold deteriorating bonds until prices have already fallen. The economic problem is therefore not simply whether ratings are inflated; it is how the rating agency trades off false positives against false negatives and delayed warnings.

7.3 Takeaway

The paper provides a clear empirical estimate of the cost of a stable, issuer-paid rating system. Ratings stability reduces volatility, but the delayed downgrade of defaulting bonds creates meaningful investor losses. The evidence supports using independent and subscriber-paid information sources as benchmarks for regulatory and contractual systems that rely on credit ratings.